```python
# ============================================================
# PYTHON MINI PROJECT
# CINEMA SEAT BOOKING SYSTEM
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# Total number of seats
TOTAL_SEATS = 20

# Dictionary to store booking information
# Key   = Seat Number
# Value = Customer details
bookings = {}


# ============================================================
# 1. VIEW SEATS
# ============================================================
def view_seats():
    print("\n========== SEAT MAP ==========")

    for seat in range(1, TOTAL_SEATS + 1):

        # X means the seat is booked
        if seat in bookings:
            print("[X]", end=" ")
        else:
            print(f"[{seat}]", end=" ")

        # Show 5 seats in each row
        if seat % 5 == 0:
            print()

    print("\nX = Booked")


# ============================================================
# 2. BOOK SEAT
# ============================================================
def book_seat():
    print("\n========== BOOK SEAT ==========")

    # Get seat number
    try:
        seat = int(input("Enter seat number: ").strip())
    except ValueError:
        print("Invalid input! Please enter a number.")
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        return

    # Check valid seat range
    if seat < 1 or seat > TOTAL_SEATS:
        print(f"Invalid seat number! Please select a seat from 1 to
{TOTAL_SEATS}.")
        return

    # Check if seat is already booked
    if seat in bookings:
        print(f"Seat {seat} is already booked.")
        print("Please select another seat.")
        return

    # Get customer name
    name = input("Enter customer name: ").strip()

    # Check empty name
    if name == "":
        print("Customer name cannot be empty.")
        return

    # Get phone number
    phone = input("Enter phone number: ").strip()

    # Basic phone validation
    if phone == "":
        print("Phone number cannot be empty.")
        return

    # Store booking
    bookings[seat] = {
        "name": name,
        "phone": phone
    }

    print(f"Seat {seat} booked successfully.")


# ============================================================
# 3. CANCEL BOOKING
# ============================================================
def cancel_booking():
    print("\n========== CANCEL BOOKING ==========")

    # Get seat number
    try:
        seat = int(input("Enter seat number: ").strip())
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    except ValueError:
        print("Invalid input! Please enter a number.")
        return

    # Check valid seat range
    if seat < 1 or seat > TOTAL_SEATS:
        print(f"Invalid seat number! Please select a seat from 1 to
{TOTAL_SEATS}.")
        return

    # Check if booking exists
    if seat not in bookings:
        print(f"No booking exists for seat {seat}.")
        return

    # Remove booking
    del bookings[seat]

    print("Booking cancelled successfully.")


# ============================================================
# 4. VIEW ALL BOOKINGS
# ============================================================
def view_bookings():
    print("\n========== CURRENT BOOKINGS ==========")

    # Check if there are no bookings
    if not bookings:
        print("No bookings found.")
        return

    print(f"{'Seat':<10}{'Customer':<20}{'Phone'}")
    print("-" * 45)

    # Display all bookings
    for seat, details in bookings.items():
        print(
            f"{seat:<10}"
            f"{details['name']:<20}"
            f"{details['phone']}"
        )


# ============================================================
# 5. SEARCH BOOKING
# ============================================================
def search_booking():
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print("\n========== SEARCH BOOKING ==========")

print("You can search by:")
print("1. Seat Number")
print("2. Customer Name")
print("3. Phone Number")

choice = input("Enter your choice: ").strip()

# Search by seat number
if choice == "1":

    try:
        seat = int(input("Enter seat number: ").strip())
    except ValueError:
        print("Invalid input! Please enter a number.")
        return

    if seat in bookings:
        details = bookings[seat]

        print("\nBooking Found")
        print(f"Customer: {details['name']}")
        print(f"Seat: {seat}")
        print(f"Phone: {details['phone']}")
    else:
        print("No booking found for this seat.")

# Search by customer name
elif choice == "2":

    name = input("Enter customer name: ").strip()

    if name == "":
        print("Customer name cannot be empty.")
        return

    found = False

    for seat, details in bookings.items():

        if details["name"].lower() == name.lower():

            print("\nBooking Found")
            print(f"Customer: {details['name']}")
            print(f"Seat: {seat}")
            print(f"Phone: {details['phone']}")
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                found = True

        if not found:
            print("No matching booking found.")

    # Search by phone number
    elif choice == "3":

        phone = input("Enter phone number: ").strip()

        if phone == "":
            print("Phone number cannot be empty.")
            return

        found = False

        for seat, details in bookings.items():

            if details["phone"] == phone:

                print("\nBooking Found")
                print(f"Customer: {details['name']}")
                print(f"Seat: {seat}")
                print(f"Phone: {details['phone']}")

                found = True

        if not found:
            print("No matching booking found.")

    else:
        print("Invalid search option.")


# ============================================================
# MAIN FUNCTION
# ============================================================
def main():

    while True:

        print("\n")
        print("=" * 40)
        print("      CINEMA SEAT BOOKING SYSTEM")
        print("=" * 40)

        print("1. View Seats")
        print("2. Book Seat")
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        print("3. Cancel Booking")
        print("4. View Bookings")
        print("5. Search Booking")
        print("6. Exit")

        print("=" * 40)

        choice = input("Enter your choice: ").strip()

        # View seats
        if choice == "1":
            view_seats()

        # Book seat
        elif choice == "2":
            book_seat()

        # Cancel booking
        elif choice == "3":
            cancel_booking()

        # View bookings
        elif choice == "4":
            view_bookings()

        # Search booking
        elif choice == "5":
            search_booking()

        # Exit
        elif choice == "6":
            print("\nThank you for using Cinema Seat Booking System!")
            print("Program exited successfully.")
            break

        # Invalid menu option
        else:
            print("Invalid choice! Please select an option from 1 to 6.")


# ============================================================
# START PROGRAM
# ============================================================
if __name__ == "__main__":
    main()
```